Networks and Systems Management

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Motivation

Imagine you are a manager at an IT company. Your network engineers and system administrators have setup the organizational network in such a way that every team is “isolated” from every other team, needing to communicate through the networks “main hub”, which has strict security rules in place. This is the best scenario. However, those engineers and admins are getting old, and they used to **do all configuration by hand**.

Feeling the need for change, you desire a solution that lets you segment separate sub-networks for any specific team and automatically configure any services that team might need, all from the comfort of you office and without needing to bother other employees.

Problems

Complexity in Setup

Traditional enterprise setups require extensive manual configurations, leading to time-consuming and error-prone processes.

Integration Problems

Incorporating new services and plugins into existing architectures can be cumbersome and may lead to compatibility issues.

Scalability Concerns

Scaling IT infrastructure to accommodate the evolving needs of the enterprise poses significant challenges in terms of resource allocation and management.

Solutions

Our platform offers a user-friendly interface enabling enterprises to **effortlessly create and manage teams**, while being able to choose within a variety of settings to enhance various functionality.

Virtualization with Docker Compose

Leveraging Docker Compose for container orchestration provides agility and scalability to the infrastructure.

Modular Design

Our solution adopts a modular architecture, allowing seamless integration of new services and plugins without disrupting existing setups.

Various settings to choose

Users are not constrained to a fixed set of configurations, allowing them to customize the configuration for each different team.

Automatic service configuration

As an example, automatic DNS simplifies network management by dynamically assigning DNS configurations to sub-networks. Other services like required proxying might be required for security purposes.

Key Benefits

Efficiency

Streamlined operations and preset configurations **reduce setup time** and **minimize manual intervention.**

Flexibility

Enterprises can **easily customize** their infrastructure by selecting from a variety of available settings. If needed, they can also create their own custom settings.

Scalability

The modular architecture **ensures scalability**, allowing enterprises to **expand** their **infrastructure as needed**.

Reliability

Automated processes and containerization **enhance reliability** and **reduce** the **risk of system failures**

Prototype

Tech Stack

FastAPI

SVELTEKIT

mongoDB

docker

Architectural Overview [1/2]

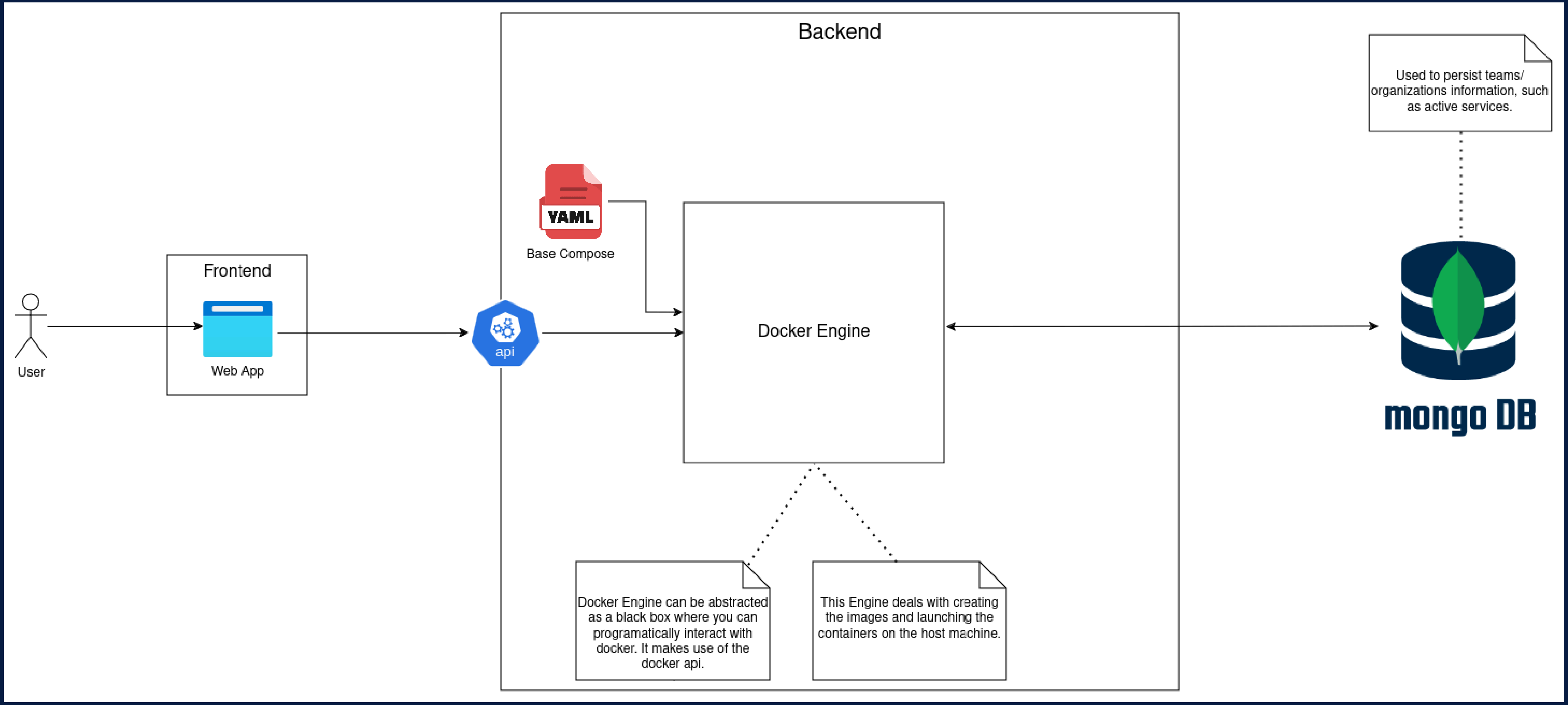

Fig. 1: Prototype's High-Level Architectural Overview

Architectural Overview [2/2]

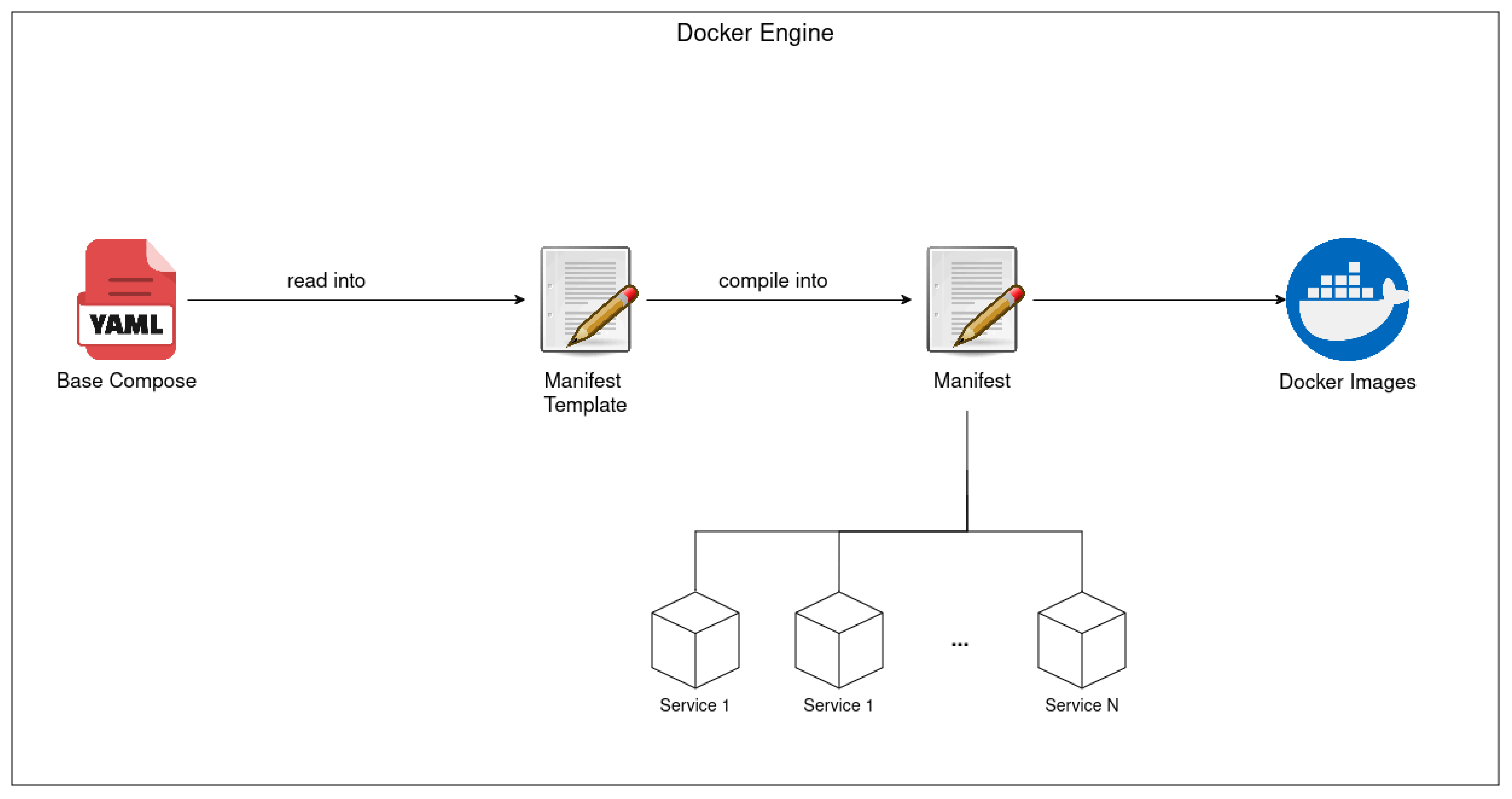

Fig. 2: Docker Engine High-Level Overview

Base Compose

```yaml
  reverse_proxy:
    image: ubuntu:20.04
    container_name: {{ teamname }}_reverse_proxy
    networks:
      {{ teamname }}_net:
        ipv4_address: {{ proxy_ip }}
    ports:
      - "3128:3128"
    cap_add:
      - NET_ADMIN
    command: >
      /bin/sh -c '
      apt update && apt install -y squid vim iproute2 iputils-ping tcpdump iptables dnsutils curl
      echo "http_port 3128
      acl localnet src {{ subnet }}
      http_access allow localnet
      http_access deny all
      visible_hostname squid
      " > /etc/squid/squid.conf &&
      echo 1 > /proc/sys/net/ipv4/ip_forward &&
      iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE &&
      iptables -t nat -A PREROUTING -i eth0 -p tcp --dport 80 -j REDIRECT --to-port 3128 &&
      iptables -t nat -A PREROUTING -i eth0 -p tcp --dport 443 -j REDIRECT --to-port 3128 &&
      squid -NYCd 1 &&
      while true; do /bin/sleep 5m; done
      '

networks:
  {{ teamname }}_net:
    driver: bridge
    ipam:
      config:
        - subnet: {{ subnet }}
```

Fig. 2: Snippet of the Base Compose

- Simple.
- Services: squid proxy, dns, web-server.
- Uses templating.
- Base for teams/orgs creation.
- Later converted to manifest to be used with the engine.

Other Services

- **Web**
  - nginx, wordpress, ...
- **CI/CD**
  - Gitlab, Jenkins, ...
- **Databases**
  - Postgres, MongoDB, Redis
- **Monitoring and Logging**
  - Prometheus, Grafana, Fluentd, Logstash
- **Messaging and Streaming**
  - RabbitMQ, Kafka, Zulip
- Others
  - Elasticsearch, squid proxy, HAProxy, Traefik, Vault, ...

Conclusions

- Could easily evolve to **support other engines**.
- Support for **more services** can be added.

Q&A + Live Demo

If you have any questions, please let us know!